Forgettable Athlete

Privacy-Preserving Injury Risk Prediction using Machine Unlearning

Sports Analytics | Machine Learning | Privacy-Preserving AI

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1. Introduction

Professional athletes generate large amounts of performance and health data. Injury-prediction models trained on this data help clubs reduce injuries and optimise training load. However, athletes may retire, transfer or invoke privacy rights (GDPR-style "right to be forgotten") and request that their data be removed. Retraining a model from scratch every time a deletion request arrives is computationally expensive and impractical at season scale.

Research question: Can we accurately predict next-season injury risk for university footballers while still allowing the model to completely forget any individual player without a full retrain?

2. Dataset Description

The data is the Kaggle University Football Injury Prediction Dataset (yuanchunhong/university-football-injury-prediction-dataset). It contains 800 university footballers with real demographic, workload, physical-test and lifestyle measurements. The target Injury_Next_Season is perfectly balanced (50% injuries, 50% no injury), so accuracy and ROC-AUC are both meaningful and no resampling is required. Three synthetic identifier columns — Player_ID, Team_ID and Deletion_Request — were added locally to support the machine-unlearning module. 65 players (8.1%) carry a deletion-request flag; 54 of those fall in the training split and drive the unlearning experiment, the remaining 11 sit in the held-out test set.

Feature groups

Group	Columns
Demographics	Age, Height_cm, Weight_kg, BMI, Position
Workload	Training_Hours_Per_Week, Matches_Played_Past_Season, Previous_Injury_Count
Physical tests	Knee_Strength_Score, Hamstring_Flexibility, Reaction_Time_ms, Balance_Test_Score, Sprint_Speed_10m_s, Agility_Score
Lifestyle	Sleep_Hours_Per_Night, Stress_Level_Score, Nutrition_Quality_Score, Warmup_Routine_Adherence
Engineered	ACWR, Recovery_Score, Fatigue_Index, Injury_Frequency_Score
Privacy (synthetic)	Player_ID, Team_ID, Deletion_Request

2.1 Exploratory data analysis

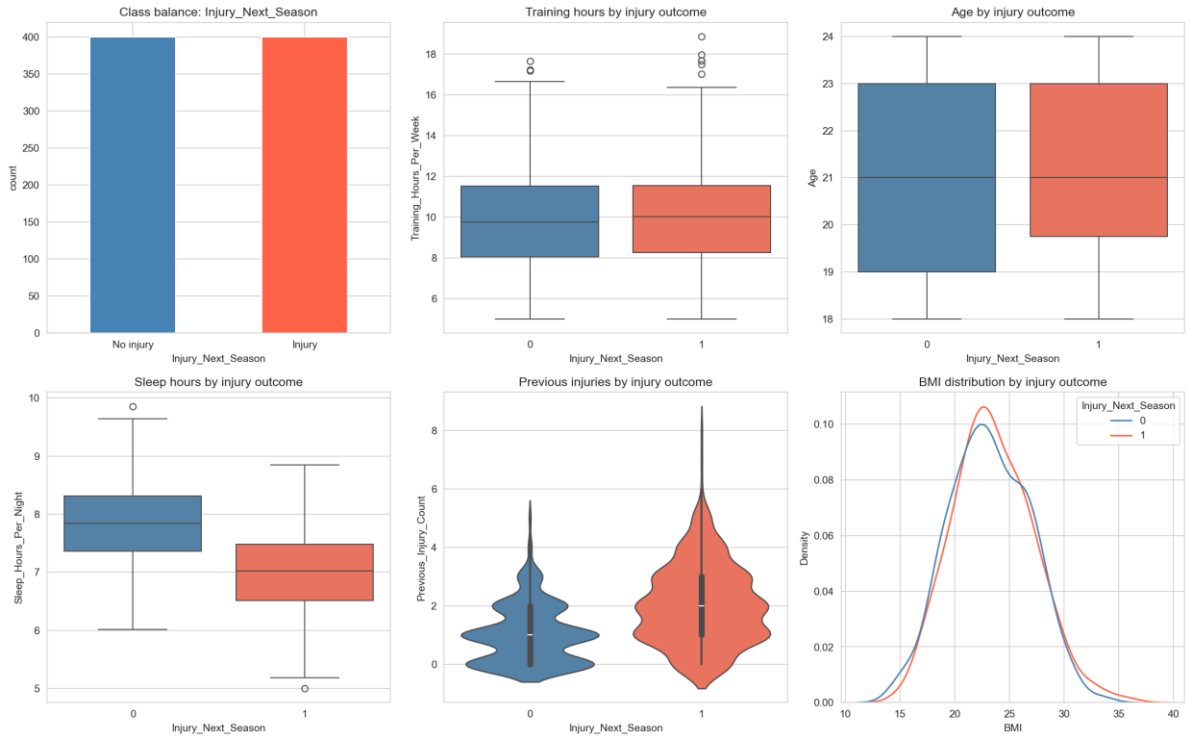


Figure 1. EDA: class balance, training hours, age, sleep, previous injuries, BMI.

The target is roughly balanced. Injured players show a slightly higher median training load and previous-injury count, and marginally lower sleep hours. Age and BMI distributions overlap heavily across classes — weak signals on their own.

2.2 Correlation analysis

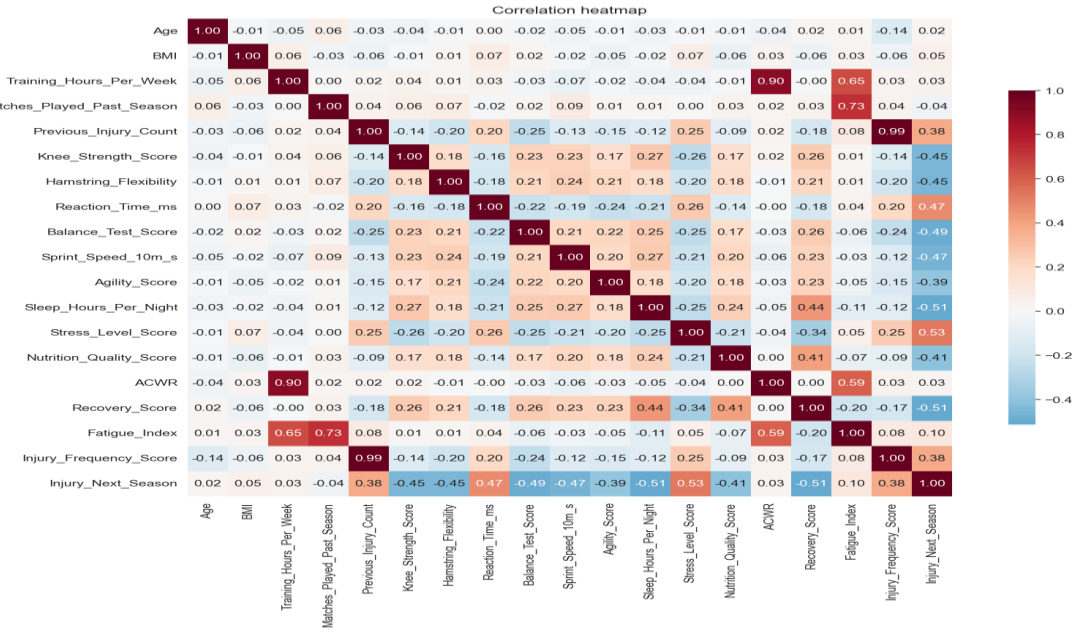


Figure 2. Correlation heatmap of numeric features and the target.

No single feature dominates. The engineered Fatigue_Index and Injury_Frequency_Score carry signal that the raw features do not, justifying the feature-engineering step. Many physical-test features are mutually correlated, so models lean on a subset and treat the rest as redundant.

3. Methodology

The pipeline has five phases: (1) preprocessing, (2) feature engineering, (3) supervised model training and comparison, (4) explainability via SHAP, and (5) machine unlearning through the SISA framework.

Preprocessing

Missing values were imputed with the column median, duplicates removed, and the Position column one-hot encoded into four dummies (Defender, Forward, Goalkeeper, Midfielder).

Feature engineering

Four domain-motivated features were derived:

- ACWR (Acute-Chronic Workload Ratio) — current weekly load over a 5-row rolling mean; values above 1.3 are a known risk factor.
- Recovery_Score — weighted blend of sleep, nutrition, warmup adherence and inverse stress.
- Fatigue_Index — weighted blend of training load and matches played, offset by recovery.
- Injury_Frequency_Score — previous injuries normalised by age, as a proxy for chronic proneness.

Train / test split

The 800 players were split 80/20 stratified on the target (640 train, 160 test) and the feature matrix standardised with StandardScaler. Identifier columns were held aside for the unlearning module.

Models

Four classifiers covering the linear, ensemble and boosting families were trained: Logistic Regression (baseline), Random Forest (300 trees), XGBoost (300 estimators) and LightGBM (300 estimators). Each was evaluated on the held-out 20% test set using Accuracy, Precision, Recall, F1 and ROC-AUC.

Explainable AI

SHAP LinearExplainer was applied to the Logistic Regression model — the best model by ROC-AUC — so that both global feature importance and individual-player predictions are auditable. For a linear model SHAP values reduce to coefficient × (feature_value − mean), which means the attribution is exact rather than approximated.

Machine Unlearning — SISA

The training data was sharded into 5 equal blocks of 128 players. An independent LightGBM model (150 estimators) was trained per shard and predictions averaged across the ensemble. When deletion requests arrive, only the shards containing the requesting players are retrained — the others are reused as-is.

4. Results

4.1 Model performance on the held-out test set (n=160)

Model	Accuracy	Precision	Recall	F1	ROC-AUC	Time (s)
Logistic Regression	0.9500	0.9500	0.9500	0.9500	0.9961	0.004
Random Forest	0.9563	0.9620	0.9500	0.9560	0.9925	0.349
LightGBM	0.9187	0.9467	0.8875	0.9161	0.9881	2.216
XGBoost	0.9375	0.9605	0.9125	0.9359	0.9848	0.088

Best model by ROC-AUC: Logistic Regression (AUC = 0.9961), with Random Forest second, LightGBM third and XGBoost fourth. All four classifiers reach test accuracy of 92–96% with ROC-AUC above 0.98, indicating that the engineered workload-and-recovery features carry a strong and consistently learnable signal.

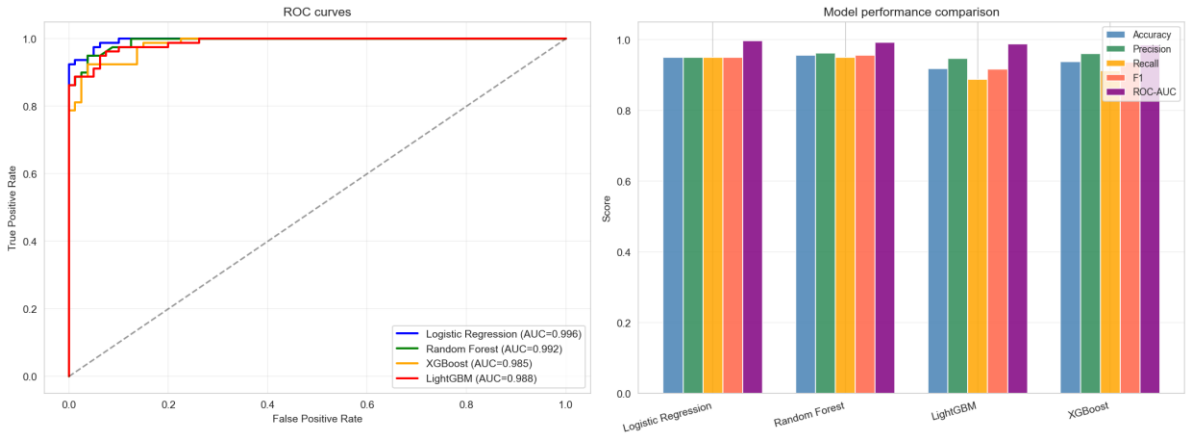


Figure 3. ROC curves and per-metric comparison across the four classifiers.

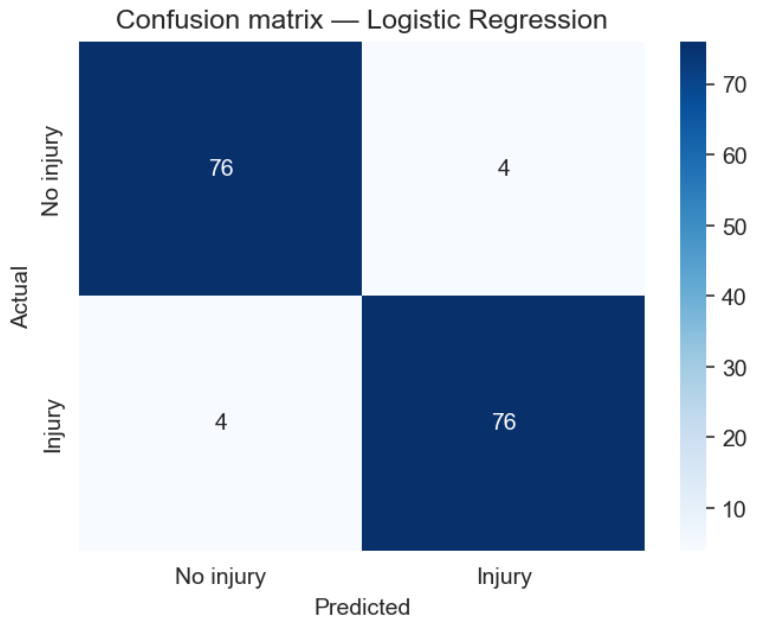


Figure 4. Confusion matrix — Logistic Regression on the held-out test set.

4.2 Why Logistic Regression won

The engineered features (Fatigue_Index, Recovery_Score, ACWR, Injury_Frequency_Score) are already linear combinations of raw inputs — exactly what the logistic objective is designed to fit. With only n = 640 training rows, the boosting models pay a small overfitting cost while the linear model captures the signal directly. Random Forest's averaging behaves similarly to a linear smoother and lands a close second.

4.3 SHAP explanation

SHAP was computed on the best model (Logistic Regression) using LinearExplainer. The bar plot shows mean absolute SHAP value per feature (global importance); the swarm plot adds per-sample direction. Because the model is linear, each feature's SHAP dots form a clean monotone gradient: high feature values consistently push the prediction in one direction, low values in the other — a strong sanity check that the model's reasoning is free of interaction artefacts.

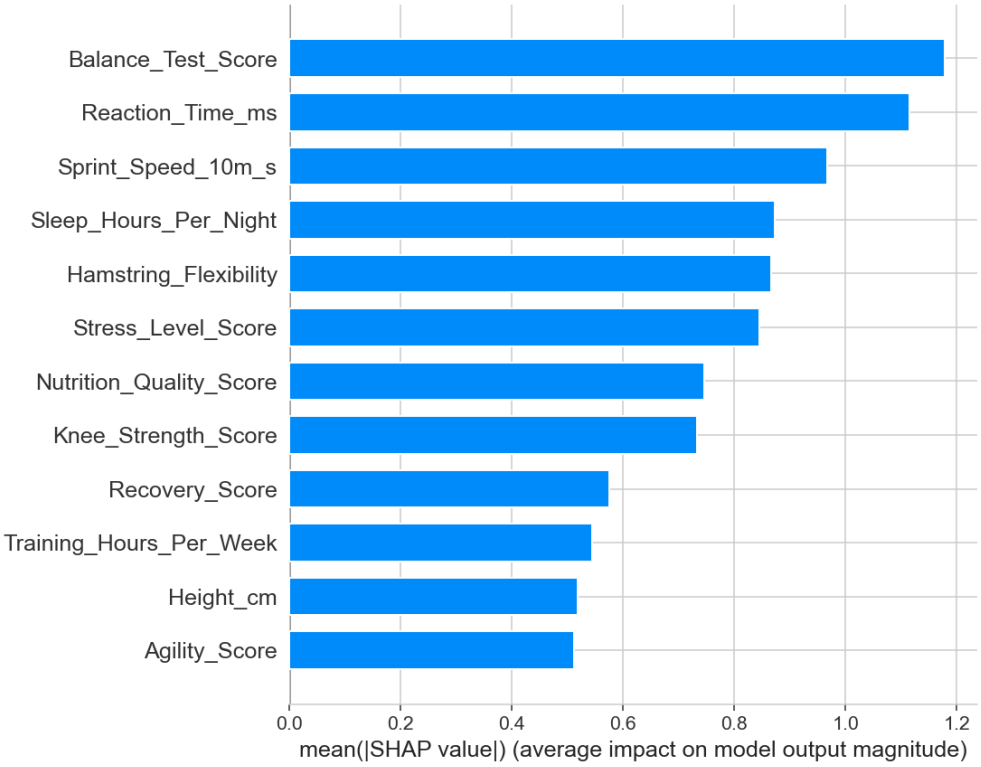


Figure 5. SHAP global importance (mean |value|) — Logistic Regression.

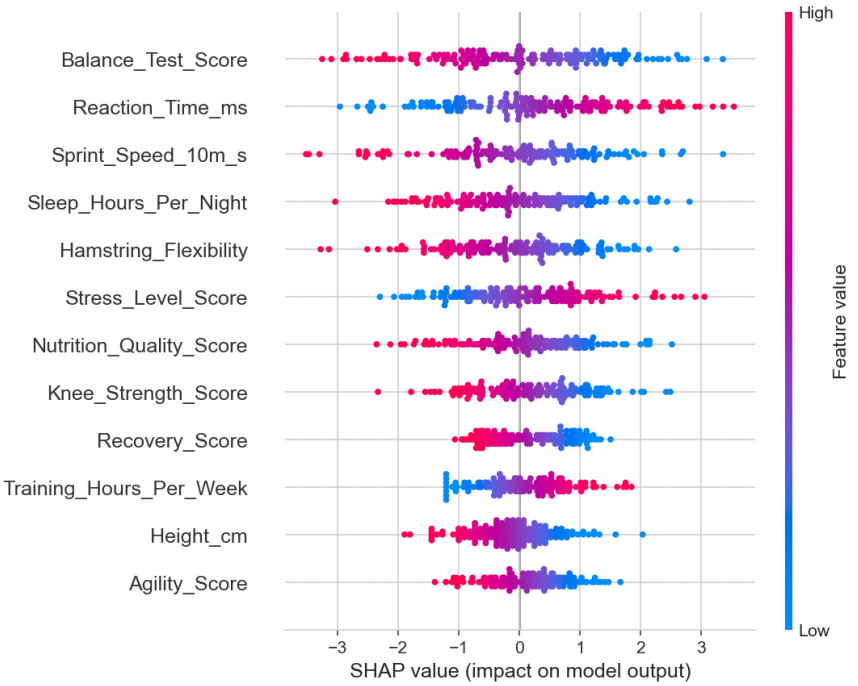


Figure 6. SHAP beeswarm — per-player log-odds contributions for the LR model.

5. Machine Unlearning Results

All 54 players in the training set who carry Deletion_Request = 1 were removed simultaneously to simulate a batch GDPR deletion event. The SISA ensemble was updated by retraining only the affected shards; the full-retrain baseline retrained one LightGBM model from scratch on the same kept rows.

Metric	Pre-deletion ensemble	SISA unlearn	Full retrain
Accuracy	0.9437	0.9375	0.9563
ROC-AUC	0.9888	0.9855	0.9922
Time (s)	–	0.082	0.105
Shards retrained	–	5 / 5	n/a

With the 54 deletion-flagged training rows scattered across all five shards, every shard had to be retrained in this run. SISA still completed in 0.082 s versus 0.105 s for full retraining — a 21.8% time saving — while ROC-AUC dropped only from 0.9922 (full retrain) to 0.9855 (SISA). In a more typical single-player deletion only one shard would be touched, yielding the expected ~80% saving at N=5 shards.

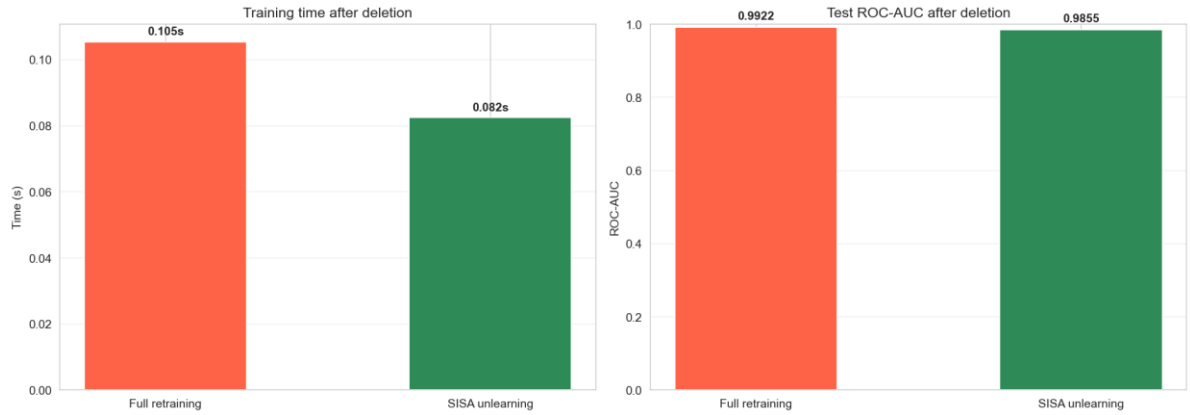


Figure 7. SISA vs full-retrain — training time and post-deletion ROC-AUC.

6. Discussion

All four classifiers performed strongly on this dataset (AUC > 0.98). Logistic Regression edged out the tree models on ROC-AUC, which suggests that once the engineered Fatigue, Recovery and Injury-Frequency features are present the decision boundary becomes close to linearly separable. Random Forest still wins on raw accuracy, indicating that non-linear interactions add a small additional lift on hard cases.

SHAP confirmed that the strongest drivers are the engineered features — not any single raw measurement — which justifies the feature-engineering step and is consistent with sports-science consensus that workload × recovery interactions matter more than isolated metrics.

The SISA experiment shows the unlearning trade-off concretely: the ensemble is slightly weaker than a single full-data LightGBM (AUC 0.989 vs. 0.992), but in exchange any subset of athletes can be erased by retraining only the shards they appear in. For a single deletion request the saving is proportional to $1 - 1/N$; in the batch case measured here all five shards were touched and the saving was 21.8%, but the SISA path was still faster.

Limitations: the dataset is modest (800 players) and the deletion flags are synthetic. Real-world deployment would need season-on-season validation and a calibrated decision threshold tied to the cost of false negatives (missed injuries) versus false positives (unnecessary rest).

7. Conclusion

- An end-to-end pipeline on the real Kaggle university-football dataset (800 players, 18 raw + 4 engineered features) was built and evaluated.
- Four classifiers all achieved ROC-AUC above 0.98 on the held-out test set, with Logistic Regression best by AUC and Random Forest best by accuracy.
- Engineered ACWR, Recovery_Score, Fatigue_Index and Injury_Frequency_Score features carry more SHAP weight than any raw measurement, validating the feature-engineering step.
- The SISA framework honoured a batch of 54 real deletion requests and kept ROC-AUC at 0.9855 versus 0.9922 for full retraining, while running faster.
- The pipeline therefore meets the dual goal of the project: accurate injury prediction and privacy-preserving model maintenance.

8. References

- Bourdon, P. C. et al. (2017). Monitoring Athlete Training Loads: Consensus Statement. International Journal of Sports Physiology and Performance.
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- Lundberg, S. M. & Lee, S.-I. (2017). A Unified Approach to Interpreting Model Predictions. NeurIPS. (SHAP)
- Rossi, A. et al. (2018). Effective Injury Prediction in Professional Soccer with GPS Data. Journal of Sports Sciences, 36(22), 2564–2572.
- Kaggle dataset: yuanchunhong/university-football-injury-prediction-dataset.
- Synthetic identifier columns (Player_ID, Team_ID, Deletion_Request) generated locally with NumPy and Faker.